

Workshop: Constrained Optimisation

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

1. Given the function:

$$z = x^2 + 2xy + 4y + 10$$

With constraint:

$$x + y = 100$$

Optimise the function z subject to the constraint:

- Write down the Lagrangian function
- Write down the three first order conditions
- Solve the first order conditions to find the optimal x , y , and the value of z .

2. Given the function:

$$z = 12x + 4y$$

With constraint:

$$x^{0.3}y^{0.7} = 1000$$

Optimise the function z subject to the constraint:

- Write down the Lagrangian function
- Write down the three first order conditions
- Solve the first order conditions to find the optimal x , y , and the value of z .

3. Given the function:

$$z = x^{0.6}y^{0.4}$$

With constraint:

$$5x + 8y = 120$$

Optimise the function z subject to the constraint